[TODO: Paper Title]

Anonymous CVPR submission

Abstract

[TODO: Abstract: summarize the problem, the proposed method, and the main results.]

1. Introduction

[TODO: Write the Introduction here: background and motivation, research gap, contributions.]

2. Method

2.1. Overview

KnowledgeLLMadapts a frozen large language model (LLM) to EEG-based classification and regression with only a small set of trainable parameters. It rests on two components. First, *knowledge-bank extraction*: the parametric knowledge of the frozen LLM is read out as compact, aspect-specific sets of vocabulary tokens that characterize the EEG views the model consumes. Second, *multimodal knowledge reprogramming*: three complementary EEG views are encoded into the LLM's token space by cross-attention over learnable fusion tokens derived from the bank **[TODO: [?]]**, so the frozen LLM can consume EEG tokens with its standard inference procedure, without fine-tuning any of its weights.

As illustrated in Fig. **??** (TODO: add figure), the pipeline consists of five stages: multi-view input representation, patch encoding, knowledge-bank extraction, knowledge reprogramming, and a frozen LLM backbone with a lightweight prediction head. Only the patch encoders, the knowledge mappers, the reprogramming layers, and the final head are trained; the LLM backbone remains frozen throughout.

2.2. Knowledge-Bank Extraction

The knowledge source of KnowledgeLLMis the frozen LLM itself. For each EEG view—equivalently, each *aspect* $a \in \{\mathrm{fc}, \mathrm{psd}, \mathrm{raw}\}$—we prepare a short statement $\mathcal{T}_a$ describing clinically established EEG knowledge of that aspect, and pass it through the *complete* frozen LLM (before layer truncation) in plain text form. From the logits of the final layer, we take the last K non-special positions; per position, special tokens are masked to $-\infty$ and the top-M tokens are retained and re-normalized with temperature τ_k. Tokens are then aggregated across positions by their probability mass and sorted, yielding the aspect bank

$$\mathcal{B}_a = \big\{(v_i, w_i)\big\}_{i=1}^{|\mathcal{B}_a|}, \tag{1}$$

i.e., a set of vocabulary token ids v_i with normalized weights w_i. Intuitively, this activation-based extraction reads out which vocabulary tokens the LLM most strongly associates with each EEG aspect, condensing the LLM's parametric knowledge into a compact, reusable token set. The bank embeddings $\mathbf{E}_a = [\mathbf{e}_{v_1}; \ldots; \mathbf{e}_{v_{|\mathcal{B}_a|}}] \in \mathbb{R}^{|\mathcal{B}_a| \times D}$ are looked up from the frozen input embedding matrix of the LLM (D is the LLM hidden size), so the bank is a projection of the LLM's knowledge onto its own token space. The extraction runs offline once and is reused throughout training; only afterwards are the first N layers of the LLM retained for the backbone, keeping the trainable pipeline light.

2.3. Multimodal Knowledge Encoding

The second component couples the bank to the EEG signal through three complementary views, each tokenized by a patch encoder and then *reprogrammed* against the bank into the LLM's hidden space.

Multi-view input representation. Given one EEG segment $\mathbf{x} \in \mathbb{R}^{C \times T}$ with C channels and T time samples, each channel is first z-scored over time to stabilize the input scale before convolution. We then build three views covering the temporal, spectral, and connectivity aspects of the signal: the normalized *raw waveform* itself; a *relative power spectral density (PSD)* per channel, estimated Welch-like with a Hann window and overlapping segments, band-limited and normalized to unit total power, $\mathbf{S} \in \mathbb{R}^{C \times F}$ with F frequency bins; and *sliding-window dynamic functional connectivity (FC)*, where the segment is split into W sliding windows, the Pearson correlation matrix $\mathbf{R}^{(w)} \in \mathbb{R}^{C \times C}$ is computed per window, and only the upper-triangular part is kept, giving an edge vector $\mathbf{e}^{(w)} \in \mathbb{R}^{E}$ with $E = C(C-1)/2$. Each view carries its own natural "sequence" axis: time, frequency, and windows.

Patch encoding. Each view is tokenized by a lightweight patch encoder with shared design [TODO: [?]]. For a view with sequence length T_m and D_m input rows, the encoder unfolds the sequence into patches of size p_m with stride s_m; replication padding is applied so that a final partial patch is preserved. The flattened patches are mapped to d-dimensional tokens by a convolutional value embedding with kernel size 3 and circular padding, followed by LayerNorm and dropout:

$$\mathbf{H}_m = g_m(\mathbf{v}_m) \in \mathbb{R}^{L_m \times d}, \qquad L_m = \left\lfloor \frac{T_m + s_m - p_m}{s_m} \right\rfloor + 1, \tag{2}$$

where $\mathbf{v}_m$ denotes the view input. We deliberately add *no* positional embedding at this stage: the frozen LLM applies its own rotary positional encoding downstream, and a fixed positional term here dominates the convolutional output at initialization, which we found leads to degenerate (collapsed) representations. The three encoders share the design but use view-specific patch geometry and are not weight-shared, producing feature tokens $\mathbf{H}_{\text{raw}}$, $\mathbf{H}_{\text{psd}}$, and $\mathbf{H}_{\text{fc}}$ of lengths L_{raw}, L_{psd}, and $L_{\text{fc}} = W$, respectively.

Knowledge reprogramming. For each aspect, the bank is compressed into P learnable *fusion tokens* by a bias-free linear mapper over the bank-token axis:

$$\mathbf{f}_a = \mathbf{W}_a \mathbf{E}_a \in \mathbb{R}^{P \times D}, \qquad \mathbf{W}_a \in \mathbb{R}^{P \times |\mathcal{B}_a|}. \tag{3}$$

The mapper is initialized by *segmented averaging*: the i-th output row averages the i-th contiguous group of bank tokens with uniform weights, so the P fusion tokens are distinct from the very start. We found this essential—a collinear initialization lets all fusion tokens collapse to the same vector early in training. Dropout is applied to the fusion tokens on the training path.

The view features are then *reprogrammed* into the LLM space by a multi-head cross-attention layer per aspect, where the feature tokens act as queries and the fusion tokens act as both keys and values. With H heads and head dimension $d_h = d/H$,

$$\mathbf{q} = \mathbf{Q}_a \mathbf{H}_a, \qquad \hat{\mathbf{k}} = \text{normalize}\big(\mathbf{K}_a \mathbf{f}_a\big), \tag{4}$$

$$\alpha_{l,p}^{(h)} = \text{softmax}_p\left(\frac{\langle \mathbf{q}_{l,h}, \hat{\mathbf{k}}_{p,h} \rangle}{\sqrt{d_h}} \right), \tag{5}$$

$$\mathbf{z}_a = \frac{1}{H} \sum_{h=1}^{H} \sum_{p=1}^{P} \alpha_{l,p}^{(h)} \mathbf{f}_{a,p} \ \in \mathbb{R}^{L_a \times D}. \tag{6}$$

The key side is unit-normalized (*scaled-cosine attention*): the frozen word-embedding scale is far smaller than the query scale, and without normalization the inter-key differences shrink to $\sim 10^{-4}$ in score space, degenerating the attention into a uniform average. Each output token $\mathbf{z}_a$ is thus a query-specific mixture of the P knowledge fusion tokens, expressed directly in the LLM's hidden space. The three aspects are concatenated into the final token sequence $\mathbf{z} \in \mathbb{R}^{L \times D}$ with total length $L = \sum_a L_a$:

$$\mathbf{z} = [\, \mathbf{z}_{\text{fc}};\ \mathbf{z}_{\text{psd}};\ \mathbf{z}_{\text{raw}} \,]. \tag{7}$$

2.4. Frozen LLM Backbone and Training Objectives

Frozen LLM backbone and prediction head. The token sequence $\mathbf{z}$ is consumed by the first N layers of the frozen LLM backbone (in our instantiation, an 8B-parameter Llama-style model [TODO: [?]]), using rotary positions $0, \ldots, L-1$ and the LLM's standard causal attention. All backbone weights are frozen; only the RMSNorm parameters of the retained layers remain trainable, following [TODO: [?]], so that the norms can absorb the scale of the injected EEG tokens and match the LLM's internal statistics:

$$\mathbf{h} = \text{LLM}_{1:N}(\mathbf{z}) \in \mathbb{R}^{L \times D}. \tag{8}$$

The representation is mean-pooled over tokens and fed into a small prediction head:

$$\hat{\mathbf{y}} = \text{Head}\Big(\text{GELU}\big(\text{LayerNorm}(\bar{\mathbf{h}})\big)\Big), \qquad \bar{\mathbf{h}} = \frac{1}{L} \sum_{l=1}^{L} \mathbf{h}_l, \tag{9}$$

with dropout before the final linear projection.

Training objectives. For classification, the primary objective is the cross-entropy loss $\mathcal{L}_{\text{CE}}(\hat{\mathbf{y}}, y)$. We add a supervised contrastive regularizer [TODO: [?]] that aligns the multi-view representations of samples sharing a label: for each aspect, the encoder tokens are mean-pooled and projected, $\mathbf{u}_a = \text{proj}(\frac{1}{L_a} \sum_l \mathbf{H}_{a,l})$, and an InfoNCE loss is computed over the batch with same-label pairs as positives (self-pairs excluded), with temperature τ_c:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda\, \mathcal{L}_{\text{contrast}}, \tag{10}$$

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \log \frac{\sum_{p \in \mathcal{P}(i)} e^{s_{i,p}/\tau_c}}{\sum_{j \neq i} e^{s_{i,j}/\tau_c}}, \tag{11}$$

where $s_{i,j}$ is the similarity between the projected view features of samples i and j, $\mathcal{P}(i)$ the positive set of i, and $\mathcal{V}$ the set of samples with a valid positive pair. This term encourages the three views to agree on the task semantics before the LLM consumes them. For regression tasks, the head is trained with the Huber loss ($\delta = 1$).

Dataset	Task (classes)	#Subj. / #Seg.
CAUEEG	AD vs. normal	**[TODO: counts]**
TUAB	abnormal vs. normal	**[TODO: counts]**
TUEP	epilepsy / non-epilepsy	**[TODO: counts]**

Table 1. Summary of the three datasets used for disease classification. **[TODO: fill subject and segment counts]**.

3. Experiments

3.1. Datasets

We evaluate on three EEG datasets covering two diseases and one electrophysiological abnormality, following the same preprocessing pipeline: all recordings are resampled to 200 Hz, selected to the standard 19-channel 10–20 montage, and re-referenced to the common average. Each recording is then cut into non-overlapping one-minute windows, and all models consume *10-second segments* of these windows (the first 10 s of each window), which is the protocol shared by the baselines below; the TimeLLMKnowledge baseline additionally follows its original 60-second-window protocol.

CAUEEG. **[TODO: [?]]** A resting-state EEG dataset of Alzheimer's disease (AD) patients and normal controls. To balance the recording length per subject, AD subjects are limited to at most 11 one-minute windows, while controls are kept uncapped.

TUAB. **[TODO: [?]]** The Temple University Abnormal EEG Corpus, with clinically labeled normal and abnormal recordings. Per-subject window caps of 3 (abnormal) and 2 (normal) are applied.

TUEP. **[TODO: [?]]** The Temple University Events Corpus, with epilepsy and non-epilepsy subjects (AR-montage sessions). Seizure segments are excluded from the continuous signal using the provided annotations before windowing, and subjects are capped at 30 (epilepsy) and 60 (non-epilepsy) windows.

For every dataset, the official train/validation/test split is stratified by subject with ratio $70/15/15$ (fixed seed 42), so no subject appears in more than one split, and all metrics are computed at the window level on the test split.

3.2. Experimental Setup

Protocol and metrics. All models are trained from scratch (no external pretraining except the model-specific initializations noted below) with 200 epochs, the Adam optimizer with learning rate 10^{-4} and weight decay 10^{-4}, a cosine learning-rate schedule, and early stopping on the validation window AUC with patience 20; the checkpoint with the best validation AUC is selected. Main-harness models are trained with DeepSpeed ZeRO-2 on 2 GPUs with a total batch size of 64; standalone baselines follow their official repositories with the same epochs, learning rate, and patience. We report window-level AUC and accuracy. Baselines are evaluated over three random seeds (42/43/44, mean $\pm$ std); TimeLLMKnowledge and KnowledgeLLMfollow the single-seed (42) archive protocol.

Evaluation modes. We consider two evaluation modes. *Same-domain*: train and test on the same dataset. *Zero-shot cross-domain*: the model trained on a source dataset is applied directly—with no fine-tuning—to the test split of a target dataset, which probes the domain-transferability of each architecture. For KnowledgeLLM, zero-shot evaluation reuses the knowledge-fusion-token snapshot stored in the checkpoint, so no re-extraction on the target domain is needed.

Baselines. We compare against five methods plus our previous reprogramming baseline: (i) **EEGNet** **[TODO: [?]]**, a compact convolutional network designed for EEG decoding; (ii) **TimeLLMKnowledge**, our earlier reprogramming baseline that encodes sliding-window dynamic functional connectivity with a GCN and feeds it into the same frozen LLM backbone (60-second windows, single seed); (iii) **NeuroLM** **[TODO: [?]]**, a language model pretrained on large-scale EEG; (iv) **LaBraM** **[TODO: [?]]**, an EEG foundation model with a neural tokenizer and vector-quantized pretraining; (v) **OFA** **[TODO: [?]]**, a one-fits-all frozen transformer adapted to EEG via input embedding patches; and (vi) **EEG-Deformer** **[TODO: [?]]**, a transformer variant for EEG with attention scaling.

3.3. Main Results

Table 2 reports same-domain and zero-shot cross-domain classification results, grouped by the training domain. All baseline cells are the mean $\pm$ standard deviation over three seeds; TimeLLMKnowledge is a single-seed archive. The KnowledgeLLMrow is reserved for the finalized runs and filled with placeholders (TODO: complete experiments and fill).

Same-domain results. **[TODO: Discussion to be finalized after the KnowledgeLLM numbers land. Observations from the table: LaBraM and TimeLLMKnowledge are the strongest same-domain models, with TimeLLMKnowledge reaching 0.8531 AUC on TUAB; EEGNet and OFA follow closely on CAUEEG.]**

Zero-shot cross-domain results. **[TODO: Discussion to be finalized. Observations: transfer out of**

Table 2. Same-domain and zero-shot cross-domain disease classification (window-level AUC / Acc). Each panel groups a training domain; columns are evaluation domains. Baselines: mean ± std over seeds 42/43/44 (10-second segments); TimeLLMKnowledge: single seed 42, 60-second windows.

	CAUEEG		TUAB		TUEP	
(a) Trained on CAUEEG	AUC	Acc	AUC	Acc	AUC	Acc
KnowledgeLLM (ours)			**[TODO: finalized runs pending]**			
TimeLLMKnowledge (ours)	0.8141	0.7224	0.7624	0.6537	0.5701	0.5626
EEGNet	0.8001±0.015	0.7408±0.009	0.6808±0.010	0.6333±0.016	0.4765±0.022	0.4717±0.038
NeuroLM	0.7606±0.017	0.6872±0.031	0.7124±0.009	0.6392±0.012	0.5153±0.013	0.5111±0.012
LaBraM	0.8244±0.006	0.7549±0.004	0.6388±0.015	0.5453±0.020	0.5737±0.022	0.4795±0.007
OFA	0.7929±0.004	0.7172±0.011	0.6825±0.008	0.6090±0.011	0.4988±0.006	0.4780±0.002
EEG-Deformer	0.8145±0.044	0.4619±0.031	0.6568±0.028	0.5347±0.002	0.4796±0.039	0.5072±0.001

	CAUEEG		TUAB		TUEP	
(b) Trained on TUAB	AUC	Acc	AUC	Acc	AUC	Acc
KnowledgeLLM (ours)			**[TODO: finalized runs pending]**			
TimeLLMKnowledge (ours)	0.6299	0.6170	0.8531	0.7812	0.4869	0.5088
EEGNet	0.5340±0.014	0.5613±0.019	0.7496±0.011	0.6101±0.047	0.5637±0.022	0.5310±0.018
NeuroLM	0.5542±0.020	0.5261±0.066	0.8422±0.009	0.7055±0.048	0.5543±0.034	0.5357±0.035
LaBraM	0.6519±0.009	0.6255±0.012	0.8332±0.005	0.7609±0.002	0.5244±0.007	0.5361±0.012
OFA	0.6160±0.011	0.5746±0.029	0.8145±0.007	0.7248±0.024	0.4516±0.002	0.5099±0.009
EEG-Deformer	0.5951±0.032	0.4966±0.091	0.8124±0.007	0.5803±0.044	0.4586±0.002	0.5006±0.036

	CAUEEG		TUAB		TUEP	
(c) Trained on TUEP	AUC	Acc	AUC	Acc	AUC	Acc
KnowledgeLLM (ours)			**[TODO: finalized runs pending]**			
TimeLLMKnowledge (ours)	0.5055	0.5051	0.5756	0.5560	0.6887	0.6203
EEGNet	0.5125±0.021	0.5587±0.001	0.4942±0.011	0.4605±0.008	0.5262±0.054	0.4994±0.010
NeuroLM	0.4579±0.036	0.5416±0.030	0.4700±0.036	0.4806±0.030	0.6331±0.017	0.5595±0.016
LaBraM	0.5122±0.012	0.5343±0.052	0.5028±0.035	0.4723±0.014	0.6349±0.013	0.5942±0.015
OFA	0.5347±0.007	0.5518±0.022	0.4825±0.035	0.4814±0.031	0.4855±0.138	0.5166±0.042
EEG-Deformer	0.5012±0.011	0.5154±0.070	0.4680±0.014	0.4892±0.041	0.6329±0.003	0.5127±0.019

CAUEEG and TUAB is substantially better than out of TUEP, where most models drop to near-chance levels; TimeLLMKnowledge achieves the strongest CAUEEG→TUAB transfer (0.7624 AUC).]

3.4. Ablation Study

[TODO: Planned: contribution of each core component (knowledge bank, multimodal reprogramming, scaled-cosine attention, segmented-average initialization, supervised contrastive loss).]

3.5. Knowledge-Bank Analysis

[TODO: Planned: effect of the knowledge texts, extraction parameters (K, M, temperature), bank size, and replacement of the bank with random or general-domain tokens.]

3.6. Reprogramming Analysis

[TODO: Planned: effect of the number of fusion tokens P, the number of retained LLM layers N, per-aspect reprogramming vs. shared, and analysis of the attention patterns over fusion tokens.]

3.7. Qualitative Analysis

[TODO: Planned: visualization of the learned fusion tokens, cross-domain representations, and error cases.]

References

[TODO: Paper Title]

Supplementary Material

4. Appendix

[TODO: Write the appendix here: additional experiments, implementation details, and supplementary analysis.]